Gunashree Rajakumar

Senior Software Engineer · Applied AI Systems · ML Infrastructure · Distributed Systems

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Boston, MA · Open to NYC · SF · Seattle · Remote

PROFESSIONAL SUMMARY

Senior Software Engineer and MS CS student at Northeastern University with 5+ years of experience designing and shipping production-grade ML infrastructure, LLM-powered agentic systems, and high-performance distributed backends across Aiera, Flipkart, Airtel, and Ola. Expertise spans RAG pipelines, LLM integration (Anthropic Claude, OpenAI), real-time feature serving, Kafka-driven event streaming, and full-stack React/TypeScript applications serving hundreds of millions of users.

EXPERIENCE

Software Engineer — ML & AI Systems
May 2025 – Dec 2025

Aiera · Boston, MA

- Built LLM inference pipelines and RAG-based hybrid retrieval (BM25 + vector search) using LangChain and OpenSearch, powering real-time financial research for hedge funds and investment banks.
- Scaled ML inference on AWS EKS to 10,000+ requests/min, reducing P95 latency by 35% through Redis caching, async pipelines, and query optimization.
- Established model observability with OpenTelemetry, Prometheus, and Grafana with A/B testing frameworks and SLA dashboards for Wall Street enterprise clients.
- Built full-stack React and TypeScript research interfaces integrating live transcripts and model outputs via enterprise-grade APIs.
- Integrated Anthropic Claude and OpenAI into production workflows for summarization and research augmentation at scale.

> ⭐ **AIERA × ANTHROPIC OFFICIAL INTEGRATION**
> Led MCP server development enabling Claude for Financial Services to access Aiera's verified market intelligence — part of Anthropic's Financial Analysis Solution for institutional investors.
> 45K+ Events/Year · 13K+ Companies · 100+ Macro Entities

Senior Software Development Engineer
Oct 2023 – Sep 2024

Flipkart · Bengaluru, India

- Architected distributed ML platform for CTR prediction, ranking, and auction optimization serving hundreds of millions of users.
- Achieved 3x throughput improvement sustaining 110+ orders/sec during Big Billion Days via Kubernetes autoscaling on Azure AKS.
- Built Spark ETL pipelines processing terabytes of user interaction data ensuring ML feature freshness.
- Reduced production incidents by 30% and root-cause resolution by 40% via Prometheus observability and distributed tracing.
- Delivered React, TypeScript, and Redux applications and shared component libraries serving hundreds of millions of users.

Senior Software Development Engineer
Jul 2022 – Oct 2023

Airtel Digital · Bengaluru, India

- Delivered sub-50ms real-time fraud detection and credit risk decisioning for Airtel Payments Bank using deep learning and GenAI patterns.
- Engineered Kafka-driven event streaming and ML feature pipelines for RBI-regulated financial decisioning with end-to-end compliance.
- Architected CI/CD across 10+ microservices using Jenkins, Docker, and Kubernetes with zero-downtime deployments.
- Built React.js dashboards for credit and fraud decisioning ML system health monitoring.
- Drove API governance across 5+ teams establishing schema contracts that eliminated integration failures.

Software Development Engineer
Aug 2020 – Jul 2022

Ola · Bengaluru, India

- Built real-time driver allocation, order orchestration, and marketplace matching across 250+ cities.
- Developed ML inference for dynamic surge pricing and driver matching with low-latency predictions.
- Engineered fault-tolerant orchestration with eventual consistency across distributed microservices.
- Optimized reliability via Redis caching, async processing, and API traffic management via Envoy Proxy.
- Built TypeScript and React.js consumer features used by millions of daily riders.

Conversational AI Intern
Jan 2020 – Jul 2020

Avaamo · Bengaluru, India

- Built NLP pipelines for Ericsson, LIC India, AB InBev — reduced production error rates by 10%.
- Raised customer satisfaction by 18% via semantic search and retrieval improvements.
- Developed TypeScript and React.js conversational AI interfaces for enterprise clients.

TECHNICAL SKILLS

LLM & GenAI: LangChain, OpenAI, Anthropic Claude, RAG, MCP Servers, Evals

ML Platform: Feature Engineering, Real-time Inference, A/B Testing, MLflow

Cloud & Infra: AWS EKS, Azure AKS, GCP, Redis, Spark, FastAPI, Spring Boot

Frontend: React.js, TypeScript, Redux, Component Libraries

Distributed Systems: Kafka, Kubernetes, Docker, Event-driven Architecture

Observability: OpenTelemetry, Prometheus, Grafana, FAISS, OpenSearch

EDUCATION

Master of Science — Computer Science
2024 – Present

Northeastern University · Boston, MA

Bachelor of Engineering — Computer Science
2016 – 2020

Global Academy of Technology · Bengaluru, India

CERTIFICATIONS

✔ Chatbot Platform Foundation ✔ Java Certification ✔ Python Certification